Bonus 4: Word Problem Toolkit

Most lost marks on Algebra 1 examinations come from word problems, and most of those come from the setup, not the algebra. This toolkit shows the ten problem types that appear most often. For each one you will find the signal words that identify it, the variables to define, the equation template, and a short model solution. Use the five-step plan from Section 4.8 with every type: read, define, write, solve, and answer in a sentence with units.

Type 1: Fixed Amount Plus a Rate

Signal words: "a fee of ... plus ... per," "starts at ... and increases by ... each." **Template:** total = fixed amount + rate × quantity.

Example B4.1 — A Repair Bill

Problem: A plumber charges a $45 call-out fee plus $30 per hour. A bill is $165. How many hours did the plumber work?

Step 1: Let h be the number of hours. Equation: $45 + 30h = 165$.

Step 2: $30h = 120$, so $h = 4$.

Answer: 4 hours. Check: $45 + 120 = 165$. ✓

Type 2: Comparing Two Plans

Signal words: "which plan is cheaper," "when will the costs be equal," "break-even." **Template:** write a model for each plan and set them equal.

Example B4.2 — Two Phone Plans

Problem: Plan A costs $20 plus $0.05 per text. Plan B costs $35 plus $0.02 per text. For how many texts do the plans cost the same?

Step 1: Let t be the number of texts. $20 + 0.05t = 35 + 0.02t$.

Step 2: $0.03t = 15$, so $t = 500$.

Answer: 500 texts (each plan costs $45). Plan A is cheaper below 500 texts and Plan B above.

Type 3: Consecutive Integers

Signal words: "consecutive," "consecutive even," "consecutive odd." **Template:** n, n + 1, n + 2 for consecutive integers; n, n + 2, n + 4 for consecutive even or odd integers.

Example B4.3 — Three Consecutive Even Integers

Problem: The sum of three consecutive even integers is 114. Find them.

Step 1: $n + (n + 2) + (n + 4) = 114$, so $3n + 6 = 114$.

Step 2: $3n = 108$, so $n = 36$.

Answer: 36, 38, and 40.

Type 4: Perimeter and Area

Signal words: "length is ... more than twice the width," "perimeter," "area." **Template:** $P = 2l + 2w$; $A = lw$. Write the length in terms of the width.

Example B4.4 — A Rectangle from Its Perimeter

Problem: A rectangle's length is 5 cm more than twice its width, and its perimeter is 64 cm. Find its dimensions.

Step 1: Width w, length $2w + 5$. Perimeter: $2(2w + 5) + 2w = 64$.

Step 2: $6w + 10 = 64$, so $w = 9$.

Answer: 9 cm by 23 cm. Check: $2(23) + 2(9) = 64$. ✓

Type 5: Distance, Rate, and Time (Wind and Current)

Signal words: "with the wind," "upstream," "downstream," "against the current." **Template:** $d = rt$, with speed $(r + w)$ in one direction and $(r - w)$ in the other.

Example B4.5 — A Boat on a River

Problem: A boat travels 60 km downstream in 3 hours and 60 km upstream in 5 hours. Find the boat's speed in still water and the speed of the current.

Step 1: Downstream speed: $60 \div 3 = 20$, so $b + c = 20$. Upstream speed: $60 \div 5 = 12$, so $b - c = 12$.

Step 2: Add: $2b = 32$, so $b = 16$ and $c = 4$.

Answer: Boat 16 km/h; current 4 km/h.

Type 6: Two Quantities, Two Totals (Tickets and Mixtures)

Signal words: "a total of ... tickets," "collected," "mixed," "blend." **Template:** one equation for the number of items and one for the total value.

Example B4.6 — Ticket Sales

Problem: A show sold 250 tickets. Adult tickets cost \$10 and student tickets cost \$6, and sales totaled \$1,900. How many of each were sold?

Step 1: $a + s = 250$ and $10a + 6s = 1900$.

Step 2: Substitute $s = 250 - a$: $10a + 1500 - 6a = 1900$, so $4a = 400$.

Answer: 100 adult and 150 student tickets.

Type 7: Percent Growth and Decay

Signal words: "increases by ...% per year," "loses ...% of its value," "interest compounded annually." **Template:** $y = a(1 + r)^t$ for growth and $y = a(1 - r)^t$ for decay.

Example B4.7 — An Investment

Problem: \$3,000 is invested at 5% interest compounded annually. What is it worth after 6 years?

Step 1: $y = 3000(1.05)^6$.

Step 2: $1.05^6 \approx 1.340096$, so $y \approx 4020.29$.

Answer: About \$4,020.29.

Type 8: Height of a Projectile

Signal words: "thrown," "launched," "height after t seconds," "hits the ground." **Template:** $h = -16t^2 + vt + h_0$ in feet (or $-4.9t^2$ in meters). Maximum: the vertex. Landing: solve $h = 0$.

Example B4.8 — A Thrown Ball

Problem: A ball is thrown upward at 48 ft/s from a height of 6 ft, so $h = -16t^2 + 48t + 6$. Find its maximum height and when it lands.

Step 1: Vertex: $t = -48/(2(-16)) = 1.5$; $h(1.5) = -36 + 72 + 6 = 42$.

Step 2: Landing: $t = (-48 - \sqrt{(48^2 + 4 \cdot 16 \cdot 6)})/(-32) = (48 + \sqrt{2688})/32 \approx 3.12$.

Answer: Maximum height 42 ft after 1.5 s; it lands after about 3.12 s.

Type 9: Maximum Area or Revenue

Signal words: "greatest area," "maximize revenue," "what price gives the most." **Template:** write the quantity as a product, expand to a quadratic, and find the vertex.

Example B4.9 — The Largest Pen

Problem: A farmer has 80 m of fencing for a rectangular pen. What dimensions give the greatest area?

Step 1: Length + width = 40, so the area is $A = x(40 - x) = -x^2 + 40x$.

Step 2: Vertex: $x = -40/(2(-1)) = 20$.

Answer: A 20 m by 20 m square, with area 400 m^2.

Type 10: Limits and Requirements (Inequalities)

Signal words: "at most," "at least," "no more than," "a minimum of," "budget." **Template:** one inequality per limit, plus $x \geq 0$ and $y \geq 0$ for real quantities.

Example B4.10 — A Snack Sale

Problem: A club sells chips for \$2 and cookies for \$3. It can sell at most 60 items and must raise at least \$150. If it sells exactly 60 items, what is the least number of cookies it can sell?

Step 1: Constraints: $c + k \le 60$, $2c + 3k \ge 150$, $c \ge 0$, $k \ge 0$.

Step 2: With 60 items, $c = 60 - k$: $2(60 - k) + 3k \ge 150$, so $120 + k \ge 150$ and $k \ge 30$.

Answer: At least 30 cookies (with 30 bags of chips, sales are exactly \$150).